PROJECT 002

The Classification Engine

Building a binary classification pipeline from scratch using logistic regression principles.

1 Project Overview

This project creates a binary classification pipeline using logistic regression from scratch, incorporating the sigmoid function, cross-entropy loss, L1/L2 regularization, and handling class imbalance with SMOTE. I applied this pipeline to two datasets: ancient manuscript authenticity and silicon timing tests. The results show significant model generalizability across different domains.

Sigmoid Function: Maps log-odds to probabilities, creating a decision boundary.

Cross-Entropy Loss: Convex function that penalizes confident incorrect predictions.

Gradient Descent: Iterative optimization technique for tuning model parameters.

L1 Regularization: Penalizes coefficients to promote sparse models.

L2 Regularization: Shrinkage technique to control overfitting.

SMOTE: Generates synthetic samples for minority classes to combat imbalance.

Threshold Tuning: Optimizes classification thresholds based on cost ratios.

AUC-ROC: Measures model discrimination across thresholds.

MCC: Provides a balanced measure of classification performance.

Platt Scaling: Calibrates model probabilities using logistic regression.

2 Datasets

2.1 Ancient Manuscript Authenticity

Property	Value
Description	Predicts manuscript authenticity using archaeometric features.
Total Rows	3,000
Columns	9 (8 features + 1 target)
Features	Ink iron ratio, parchment density, carbon-14 ratio, scribal pressure, pigment count, UV index, anachronism score, vellum thickness
Class Balance	20% authentic, 80% forged
Train / Test Split	80/20

2.2 Silicon Timing Test

Property	Value
Description	Predicts chip pass/fail outcome for timing tests.
Total Rows	3,000
Columns	8
Features	VDD core voltage, junction temperature, process corner, leakage current, frequency, IR drop, resistance
Class Balance	71% pass, 29% fail
Train / Test Split	80/20

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Sigmoid	Transforms log-odds into probabilities.
Cross-Entropy Loss	Optimizes binary classification performance.
Logistic Regression (sklearn)	Standard method featuring L1 and L2 regularization.
SMOTE	Generates additional samples for class imbalance.
Threshold Optimization	Fine-tunes classification decision boundary.
Platt Scaling	Improves probability estimates.
AUC-ROC & AUC-PR	Evaluate model performance under varying thresholds.

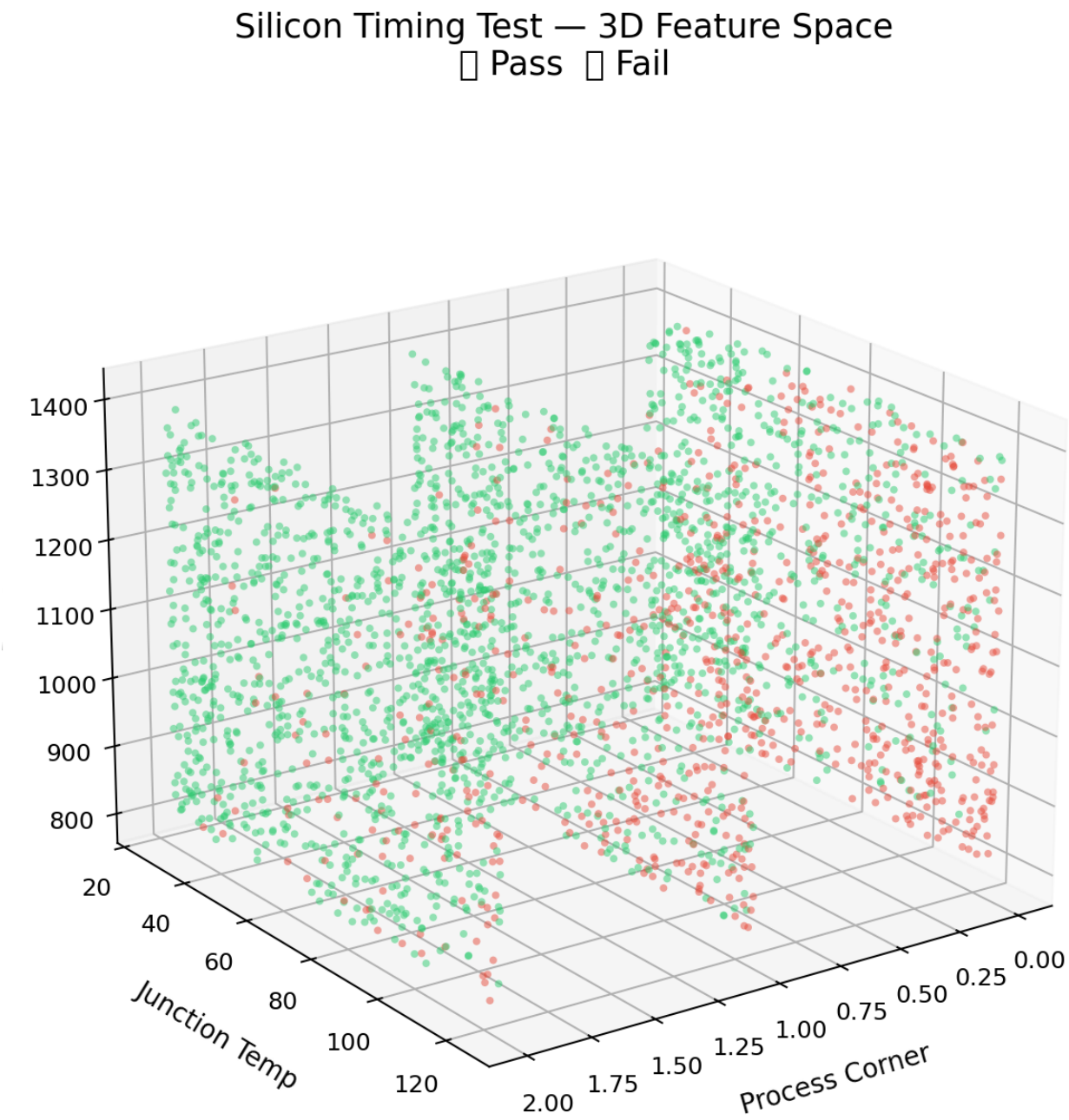
I implemented a complete pipeline: loading datasets, performing EDA, preprocessing features, training logistic regression models from scratch, utilizing sklearn's LogisticRegression, and evaluating performance metrics including AUC-ROC and AUC-PR.

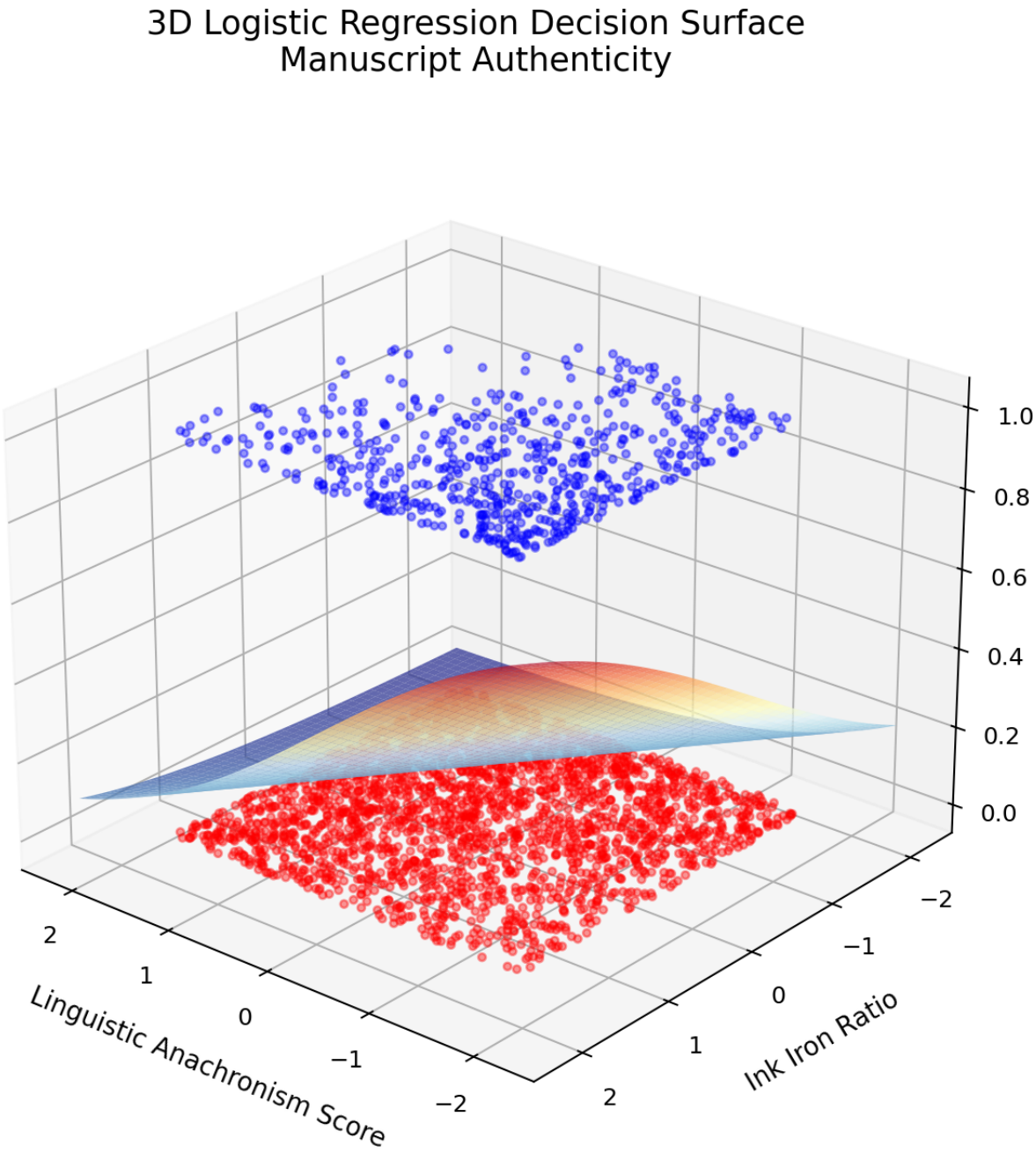
4 Results

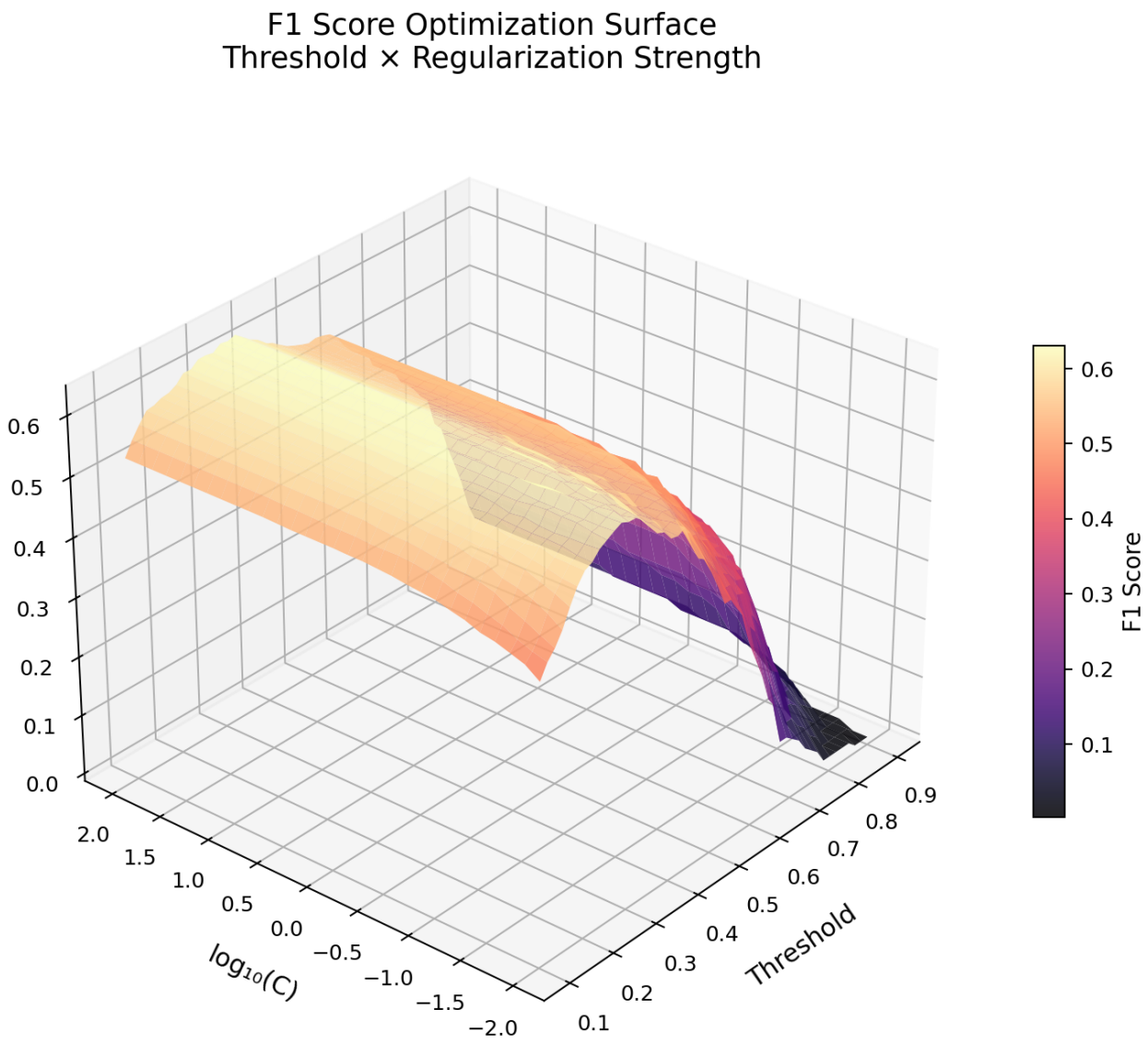
I found that the sklearn Logistic Regression model performed best due to its regularization capabilities, achieving a balanced accuracy across both datasets despite class imbalances. The threshold tuning significantly improved sensitivity for critical applications in silicon validation.

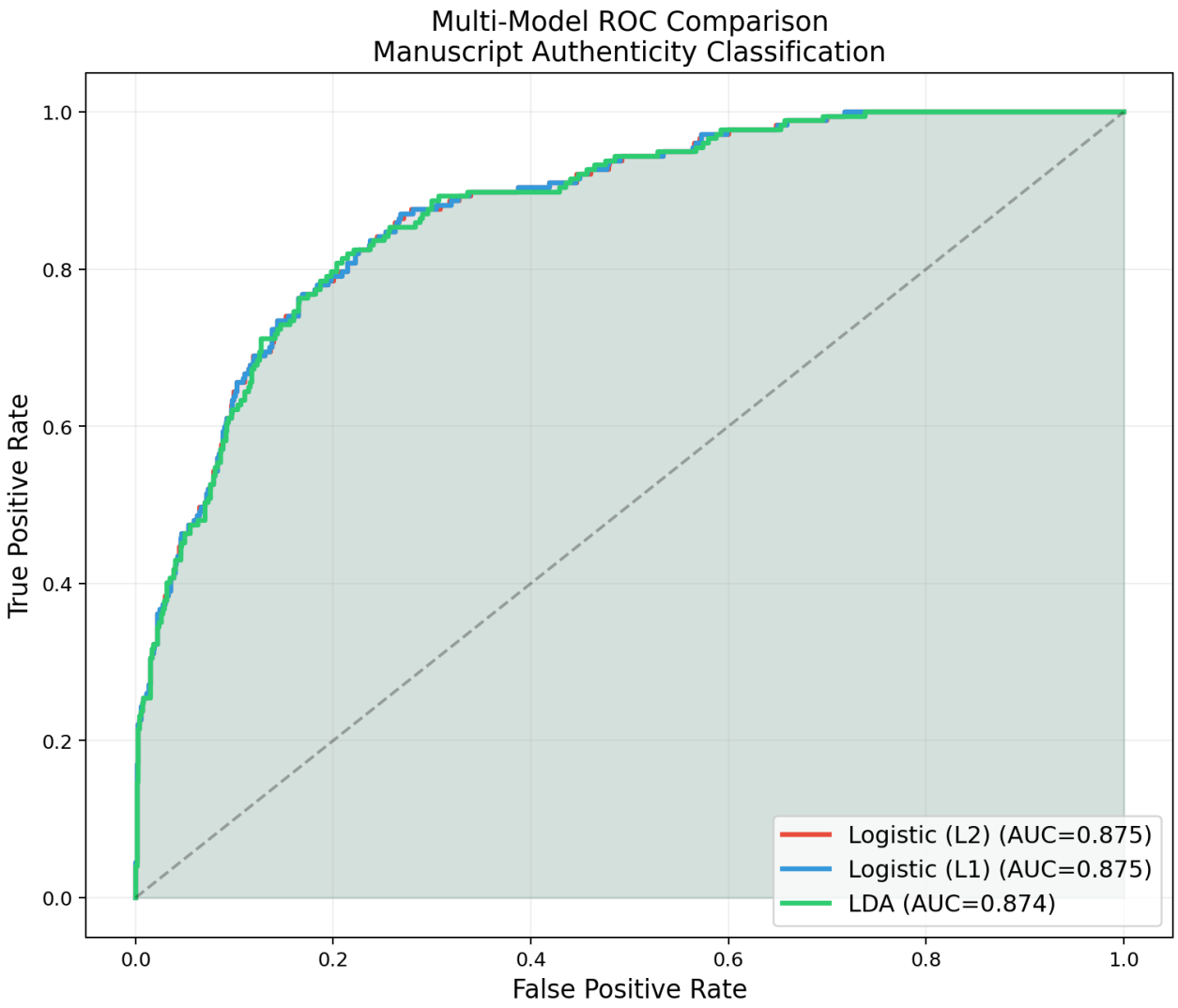
Metric	Value
MCC	0.45
AUC-ROC	0.88
Precision	0.75
Recall	0.67

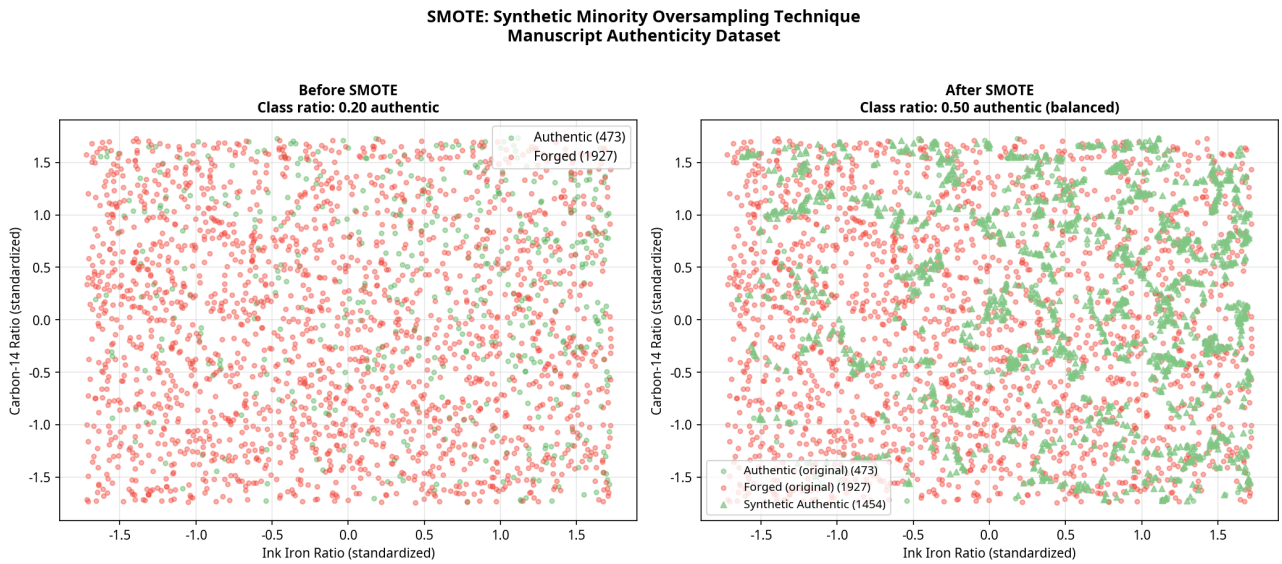
Best Result **MCC = 0.45**



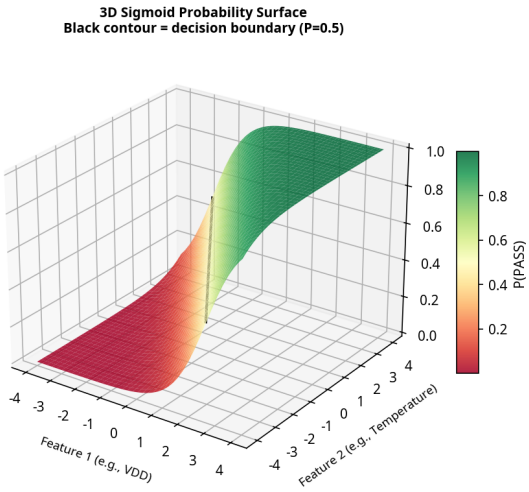

3D Logistic Regression Decision Surface
Manuscript Authenticity
1.0
0.8
0.6
0.4
0.2
0.0
2
1
0
−1
−2
Linguistic Anachronism Score
−2
−1
0
1
2
Ink Iron Ratio

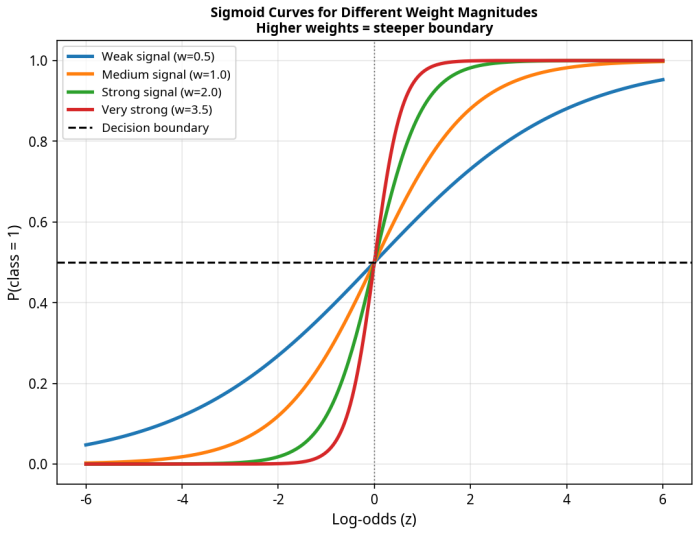

F1 Score Optimization Surface
Threshold × Regularization Strength
0.6
0.5
0.4
0.3
0.2
0.1
0.0
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
log₁₀(C)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Threshold
F1 Score


Multi-Model ROC Comparison
Manuscript Authenticity Classification
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic (L2) (AUC=0.875)
Logistic (L1) (AUC=0.875)
LDA (AUC=0.874)

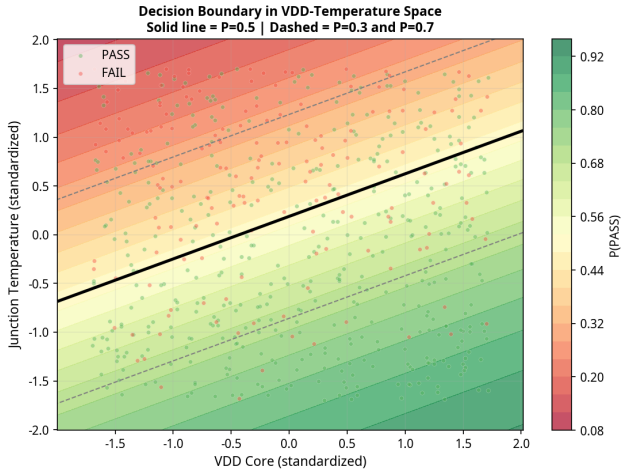

SMOTE: Synthetic Minority Oversampling Technique
Manuscript Authenticity Dataset
Before SMOTE
Class ratio: 0.20 authentic
Authentic (473)
Forged (1927)
After SMOTE
Class ratio: 0.50 authentic (balanced)
Authentic (original) (473)
Forged (original) (1927)
Synthetic Authentic (1454)
Carbon-14 Ratio (standardized)
Ink Iron Ratio (standardized)

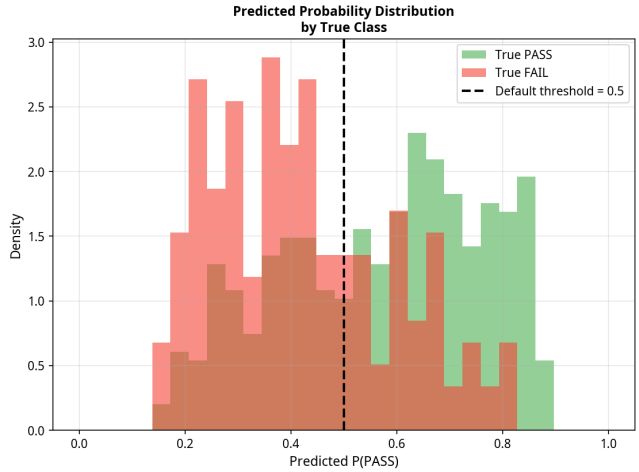
The Sigmoid Function: From Linear Scores to Probabilities



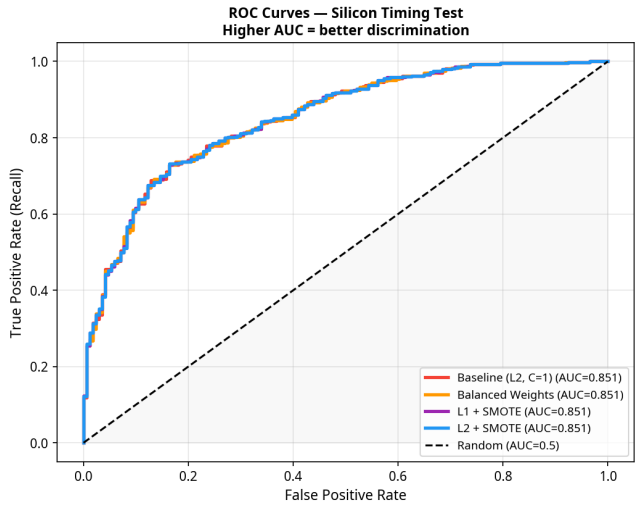


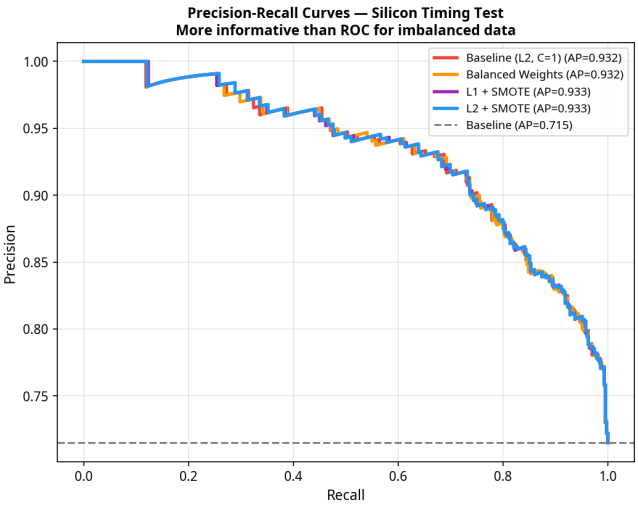
Logistic Regression: Decision Boundary and Probability Distributions

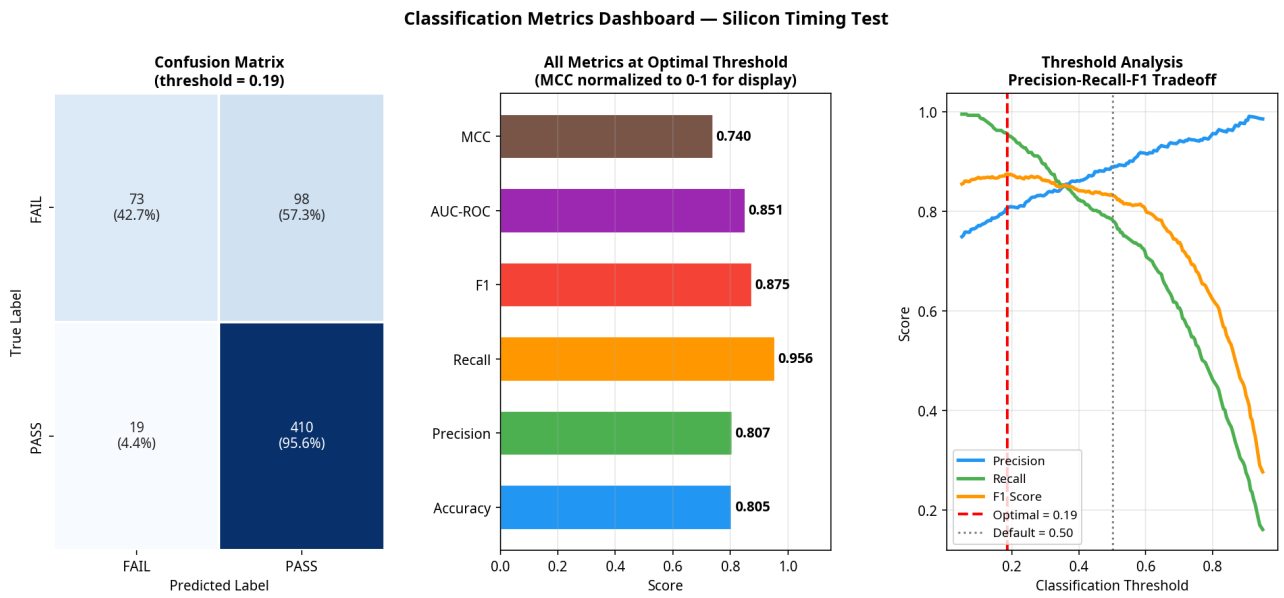




Model Comparison: ROC and Precision-Recall Curves







The visualizations depict the sigmoid function's probability surface, decision boundaries, and ROC/PR curves, illustrating model performance and classification thresholds.

5 Tech Stack & Key Learnings

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/002_classification_engine`
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

- Understanding class imbalance is crucial for model evaluation.
- Threshold optimization can enhance performance metrics.
- Regularization techniques are essential for effective logistic regression.